Statistical Signal Processing
Lecture 10:
System Identification and Frequency Response Functions

Literature and Problems: See course schedule

Agenda (preliminary):

SLIDE: Non-parametric system identification – intro

- **What, why and how ?**
- **Review of correlation functions, PSD's and coherence function**

Ideal stochastic system identification (no input or output measurement noise)

- **The H_1 and H_2 estimators**
- **Coherence function definition and $\gamma_{xy}^2(f) = \frac{\hat{H}_1(f)}{\hat{H}_2(f)} = 1$ in the ideal case**
- **DEMO: examples**

Model with extraneous input and output measurement noise

- **Signals and noise uncorrelated => PSD's are additive at input and output**

Case with only output measurement noise (most common situation)

- **Optimal estimator is H_1 (is also LS optimal, i.e. gives minimal $G_{nn}(f)$)**
- **Bias of H_1 => $|H_1| < |H|$**
- **Coherence function expressed in $SNR_o = G_{vv}/G_{nn}$ at output**
- **DEMO: examples**

Case with only input measurement noise

- **Optimal estimator is H_2 (is also LS optimal, i.e. gives minimal $G_{mm}(f)$)**
- **Bias of H_2 => $|H_2| > |H|$, same phase as H_1**
- **Coherence function expressed in $SNR_i = G_{uu}/G_{mm}$ at input**
- **DEMO: examples**

Case with input AND output measurement noise

- **Optimal estimator ?**
- **Turns out not to be possible to find without knowledge of either G_{mm} or G_{nn}**
- **Remedies: 1) assume either input or output noise to be dominant**
  2) use geometric mean $\hat{H}(f) = \sqrt{\hat{H}_1(f)\hat{H}_2(f)}$
  (equivalent to take average of $\hat{H}_1(f)$ and $\hat{H}_2(f)$ expressed in dB)
- **Coherence function expressed in $SNR_i = G_{uu}/G_{mm}$ and $SNR_o = G_{vv}/G_{nn}$**
- **DEMO: examples**

Estimation errors:

- **Model errors**
- **Estimator errors (bias error vs. random error trade-off for fixed signal length)**

SLIDE: Non-parametric system identification of MIMO systems – intro